

Exercise 4: Functions

Aim

To implement MySQL functions for banking operations such as calculating a customer's age, computing monthly loan installments, and checking account balance before transactions.

--Utilizing the same database from exercise 1 and 3

--**Question:** Write a function CalculateAge that takes a customer's date of birth as input and returns their age in years.

```
ALTER TABLE Customer
```

```
ADD DOB DATE;
```

```
UPDATE Customer SET DOB='1961-05-15' WHERE CustomerID=101;
```

```
UPDATE Customer SET DOB='1981-09-20' WHERE CustomerID=102;
```

```
UPDATE Customer SET DOB='1956-12-10' WHERE CustomerID=103;
```

```
UPDATE Customer SET DOB='1991-07-08' WHERE CustomerID=104;
```

```
UPDATE Customer SET DOB='1964-03-25' WHERE CustomerID=105;
```

```
DELIMITER $$
```

```
CREATE FUNCTION CalculateAge(
```

```
    DOB DATE
```

```
)
```

```
RETURNS INT
```

```
DETERMINISTIC
```

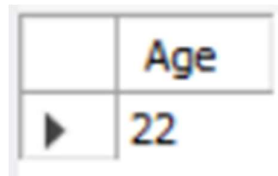
```
BEGIN
```

```
    RETURN TIMESTAMPDIFF(YEAR, DOB, CURDATE());
```

```
END $$
```

```
DELIMITER ;
```

```
SELECT CalculateAge('2004-06-15') AS Age;
```



	Age
▶	22

--**Question:** Write a function **CalculateMonthlyInstallment** that takes the loan amount, interest rate, and loan duration in years as input and returns the monthly installment amount.

```
DELIMITER $$
```

```
CREATE FUNCTION CalculateMonthlyInstallment(
```

```
    LoanAmount DECIMAL(10,2),
```

```
    InterestRate DECIMAL(5,2),
```

```
    Years INT
```

```
)
```

```
RETURNS DECIMAL(10,2)
```

```
DETERMINISTIC
```

```
BEGIN
```

```
    DECLARE EMI DECIMAL(10,2);
```

```
    SET EMI =
```

```
    (LoanAmount +
```

```
    (LoanAmount * InterestRate * Years /100))
```

```
    /(Years*12);
```

```
    RETURN EMI;
```

```
END $$
```

```
DELIMITER ;
```

```
SELECT CalculateMonthlyInstallment(500000,10,5) AS MonthlyInstallment;
```

	MonthlyInstallment
▶	12500.00

--**Question:** Write a function **HasSufficientBalance** that takes an account ID and an amount as input and returns a boolean indicating whether the account has at least the specified amount.

```
DELIMITER $$
```

```
CREATE FUNCTION HasSufficientBalance(
```

AccID INT,

Amount DECIMAL(10,2)

)

RETURNS BOOLEAN

DETERMINISTIC

BEGIN

DECLARE CurrentBalance DECIMAL(10,2);

SELECT Balance
INTO CurrentBalance
FROM Account
WHERE AccountID = AccID;

RETURN CurrentBalance >= Amount;

END \$\$

DELIMITER ;

SELECT HasSufficientBalance(1,5000) AS Status;

	Status
▶	1

SELECT HasSufficientBalance(1,500000) AS Status;

	Status
▶	0